# Anand Bhattad

Pixels · Perception · Physics

Assistant Professor of Computer Science · DSAI & Institute of Robotics
Johns Hopkins University

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## Research

Generative models look right without being right. My 3P Vision group builds the tools to tell the difference — to read out what they know and don't, and how to improve them — along three Ps:

**Pixels** — controllable generative models for relighting, compositing, and editing.

**Perception** — reading out the scene structure models implicitly know: intrinsics, geometry, and parts.

**Physics** — testing and enforcing physical and projective consistency, and carrying it to robots.

The group looks beyond generative models too: vision architectures, 3D perception, multisensory perception for embodied AI, and generative models for computational mechanics.

Keywords: Visual Generative Models; 3D Vision; Vision-based Robotics; Neural Rendering; Inverse Graphics; Intrinsic Images; Image-based Lighting; Physics Awareness in Large Vision Models.

## Education

**University of Illinois Urbana-Champaign (UIUC)**, USA

Ph.D., Computer Science 2019 – 2024

Thesis: **Exploring Knowledge in Generative Models**

Thesis committee: **David Forsyth (Advisor)**, Alexei A. Efros, William T. Freeman, Derek Hoiem, Svetlana Lazebnik, Shenlong Wang

M.S., Computer Science 2017 – 2018

M.S., Civil Engineering 2015 – 2017

**National Institute of Technology Karnataka (NITK) Surathkal**, India

B.Tech., Civil Engineering 2011 – 2015

## Work Experience

**Johns Hopkins University**, Assistant Professor of Computer Science 2025 – present
**Toyota Technological Institute at Chicago**, Research Assistant Professor 2023 – 2025
**University of California Berkeley**, Visiting Scholar; Host: Alexei A. Efros 2024
**Allen Institute for AI (Ai2)**, Research Intern with Aniruddha Kembhavi 2023
**Intel**, Research Intern with Stephan R. Richter in Vladlen Koltun's team 2021
**NVIDIA**, Research Intern with Andrew Tao and Bryan Catanzaro 2020
**Fyusion Inc**, Research Intern with Abhishek Kar and Rodrigo Ortiz Cayon 2019
**Siemens Corporate Technology**, Computer Vision & DL Intern with Jan Ernst 2017
**Michigan State University**, Research Intern with Nizar Lajnef 2014
**IIIT Hyderabad**, Research Intern with Ramacharla Pradeep Kumar 2013

## Selected Honors and Awards

**Best Paper Award in Robot Learning**, ICRA 2026
**Outstanding Reviewer**, ICCV 2023
**Best Paper Finalist**, CVPR 2022
**Outstanding Emergency Reviewer**, CVPR 2021
**Excellent Teaching Assistant**, Intro Computing: Engrg & Sci 2016

## Selected Publications

*Self-selected publications (not necessarily the most cited work).*

- OpenTac: An Open Ecosystem for Customizable, Scalable, and Transferable 3D Tactile Sensing
  **on arXiv**
- Generative Cinematographer: Composing Camera and Object Motion in 3D
  **on arXiv**
- PowerSim: Differentiable Physics Simulation and Rendering with Power Diagrams
  **on arXiv**
- Shadows Don't Lie and Lines Can't Bend! Generative Models Don't Know Projective Geometry...For Now
  **CVPR 2024**
- StyleGAN Knows Normals, Depth, Albedo, and More
  **NeurIPS 2023**
- PRINCIPIA: Relational Physics Tests for Video Models
  **NeurIPS 2026 (Evaluations & Datasets Track)**
- Visual Jenga: Discovering Object Dependencies via Counterfactual Inpainting
  **NeurIPS 2025**
- Latent Intrinsics Emerge from Training to Relight
  **NeurIPS 2024** (**Spotlight**)
- Generative Blocks World: Moving Things Around in Pictures
  **ICLR 2026**
- Do You Know Where Your Camera Is? View-Invariant Policy Learning with Camera Conditioning
  **ICRA 2026** (**Best Paper Award in Robot Learning**)
- LumiNet: Latent Intrinsics Meets Diffusion Models for Indoor Scene Relighting
  **CVPR 2025**

## Papers Under Review

⋆ denotes equal contribution, † indicates equal advising, and underlined authors are students I advised or co-mentored

6. **OpenTac: An Open Ecosystem for Customizable, Scalable, and Transferable 3D Tactile Sensing**
   Ariel Slepyan, Geeta Chandra Raju Bethala, Jiafeng Gu, Anagha Rajendra Dangle, Yinzhe Zhou, Hongyuan Fu, Jade Sim, Tuhin Basu, Xinyi Yin, Md Faizal Karim, Gregory Hager†, Homanga Bharadhwaj†, **Anand Bhattad**†, Krishna Murthy Jatavallabhula†, Nitish V. Thakor†
   **on arXiv**
   Paper Project Code
5. **OmniRay: Camera-Agnostic Monocular 3D Reconstruction with Composite Ray Fields**
   Namitha Guruprasad, Abhay Yadav, Cheng Peng, Rama Chellappa†, **Anand Bhattad**†
   **on arXiv**
   Paper Project

4. **Generative Cinematographer: Composing Camera and Object Motion in 3D**
   Jiahan Zhang, Chaohao Yang, Namitha Guruprasad, Vivekjyoti Banerjee, Trong-Tung Nguyen, Alan Yuille, **Anand Bhattad**
   **on arXiv**
   Paper Project Code
3. **PowerSim: Differentiable Physics Simulation and Rendering with Power Diagrams**
   Trong-Tung Nguyen, **Anand Bhattad**
   **on arXiv**
   Paper Project Code
2. **FracGen: Learning How Objects Stretch and Tear with Physics-Informed Video Generation**
   Trong-Tung Nguyen, Jiahan Zhang, **Anand Bhattad**
   **on arXiv**
   Paper Project
1. **ViT-5: Vision Transformers for The Mid-2020s**
   Feng Wang, Sucheng Ren, Tiezheng Zhang, Predrag Neskovic, **Anand Bhattad**, Cihang Xie, Alan Yuille
   **on arXiv**
   Paper Code

## Refereed Conference Papers

32. **PRINCIPIA: Relational Physics Tests for Video Models**
    Varun Varma Thozhiyoor*, Shivam Tripathi*, Venkatesh Babu Radhakrishnan, **Anand Bhattad**
    **NeurIPS 2026 (Evaluations & Datasets Track)**
    Paper Project Code
31. **Thinking in Boxes: 3D Editing in Real Images Made Easy**
    Pradhaan S Bhat*, Naveen Chandra*, Rishubh Parihar, Vaibhav Vavilala, Venkatesh Babu, D.A. Forsyth, **Anand Bhattad**
    **NeurIPS 2026**
    Paper Project Code
30. **SyncLight: Single-Edit Multi-View Relighting**
    David Serrano-Lozano, **Anand Bhattad**†, Luis Herranz, Jean-François Lalonde†, Javier Vazquez-Corral†
    **NeurIPS 2026**
    Paper Project Code
29. **RATS! Patches Talk Through Registers: Emergent Parts in Register Attention Transformers**
    Timing Yang, Predrag Neskovic, Jansen Seheult, Wenchao Han, **Anand Bhattad**, Alan Yuille, Feng Wang
    **NeurIPS 2026**
    Paper Code
28. **Context-Aware Autoregressive Image Generation for Emerging Reasoning Properties**
    Jixuan Ying, Haoyu Liu, Timing Yang, Tingyu Zhu, Predrag Neskovic, **Anand Bhattad**, Cihang Xie, Zeyu Zheng, Alan Yuille, Feng Wang
    **NeurIPS 2026**
27. **SyncFix: Fixing 3D Reconstructions via Multi-View Synchronization**
    Deming Li, Abhay Yadav, Cheng Peng, Rama Chellappa, **Anand Bhattad**
    **ECCV 2026**
    Paper Project Code

26. **Relighting as a Probe of Visual Priors via Augmented Latent Intrinsics**
Xiaoyan Xing, Xiao Zhang, Sezer Karaoglu, Theo Gevers, **Anand Bhattad**
**ICML 2026**
Paper Project Code

25. **Objects in Generated Videos Are Slower Than They Appear: Models Suffer Sub-Earth Gravity and Don't Know Galileo's Principle...for now**
Varun Varma Thozhiyoor, Shivam Tripathi, Venkatesh Babu Radhakrishnan, **Anand Bhattad**
**CVPR 2026 (Findings)**
Paper Project Code

24. **Name That Part: 3D Part Segmentation and Naming**
Soumava Paul, Prakhar Kaushik, Ankit Vaidya, **Anand Bhattad**, Alan Yuille
**CVPR 2026 (Findings)**
Paper Project Code

23. **Generalizable Sparse-View 3D Reconstruction from Unconstrained Images**
Vinayak Gupta, Chih-Hao Lin, Shenlong Wang, **Anand Bhattad**, Jia-Bin Huang
**CVPR 2026**
Paper Project Code

22. **Do You Know Where Your Camera Is? View-Invariant Policy Learning with Camera Conditioning**
Tianchong Jiang, Jingtian Ji, Xiangshan Tan, Jiading Fang, **Anand Bhattad**, Vitor Guizilini, Matthew R. Walter
**ICRA 2026** (**Oral**; **Best Paper Award in Robot Learning**)
Paper Project Code

21. **Generative Blocks World: Moving Things Around in Pictures**
Vaibhav Vavilala, Seemandhar Jain, Rahul Vasanth, D.A. Forsyth, **Anand Bhattad**
**ICLR 2026**
Paper Project Code

20. **SpotLight: Shadow-Guided Object Relighting via Diffusion**
Frédéric Fortier-Chouinard, Zitian Zhang, Louis-Etienne Messier, Mathieu Garon, **Anand Bhattad**, Jean-François Lalonde
**3DV 2026**
Paper Project Code

19. **Improved Convex Decomposition with Ensembling and Boolean Primitives**
Vaibhav Vavilala, Florian Kluger, Seemandhar Jain, Bodo Rosenhahn, **Anand Bhattad**, D.A. Forsyth
**3DV 2026**
Paper

18. **Visual Jenga: Discovering Object Dependencies via Counterfactual Inpainting**
**Anand Bhattad**, Konpat Preechakul, Alexei A. Efros
**NeurIPS 2025**
Paper Project

17. **LumiNet: Latent Intrinsics Meets Diffusion Models for Indoor Scene Relighting**
Xiaoyan Xing, Konrad Groh, Sezer Karaoglu, Theo Gevers, **Anand Bhattad**
**CVPR 2025**
Paper Project Code

16. **ScribbleLight: Single Image Indoor Relighting with Scribbles**
Jun Myeong Choi, Annie N. Wang, Pieter Peers, **Anand Bhattad**, Roni Sengupta
**CVPR 2025**
Paper Project Code

15. **UrbanIR: Large-Scale Urban Scene Inverse Rendering from a Single Video**
Zhi-Hao Lin, Bohan Liu, Yi-Ting Chen, Kuan-Sheng Chen, D.A. Forsyth, Jia-Bin Huang, **Anand Bhattad**, Shenlong Wang
**3DV 2025**
Paper Project Code

14. **ZeroComp: Zero-shot Object Compositing from Image Intrinsics via Diffusion**
Zitian Zhang, Frédéric Fortier-Chouinard, Mathieu Garon, **Anand Bhattad**, Jean-François Lalonde
**WACV 2025** (**Oral presentation**)
Paper Project Code

13. **Latent Intrinsics Emerge from Training to Relight**
Xiao Zhang, William Gao, Seemandhar Jain, Michael Maire, D.A. Forsyth, **Anand Bhattad**
**NeurIPS 2024** (**Spotlight presentation**)
Paper Project Code

12. **From an Image to a Scene: Learning to Imagine the World from a Million 360° Videos**
Matthew Wallingford, **Anand Bhattad**, Aditya Kusupati, Vivek Ramanujan, Matt Deitke, Sham Kakade, Aniruddha Kembhavi, Roozbeh Mottaghi, Wei-Chiu Ma, Ali Farhadi
**NeurIPS 2024**
Paper Project Code

11. **Videoshop: Localized Semantic Video Editing with Noise-Extrapolated Diffusion Inversion**
Xiang Fan, **Anand Bhattad**†, Ranjay Krishna†
**ECCV 2024**
Paper Project Code

10. ***Shadows Don't Lie and Lines Can't Bend!* Generative Models Don't Know Projective Geometry...For Now**
Ayush Sarkar*, Hanlin Mai*, Amitabh Mahapatra*, Svetlana Lazebnik, D.A. Forsyth, **Anand Bhattad**
**CVPR 2024**
Paper Project Code

9. **StyLitGAN: Image-based Relighting via Latent Control**
**Anand Bhattad**, James Soole, D.A. Forsyth
**CVPR 2024**
Paper Project Code

8. **StyleGAN Knows Normals, Depth, Albedo, and More**
**Anand Bhattad**, Daniel McKee, Derek Hoiem, D.A. Forsyth
**NeurIPS 2023**
Paper

7. **OBJect 3DIT: Language-guided 3D-aware Image Editing**
Oscar Michel, **Anand Bhattad**, Eli Vanderbilt, Ranjay Krishna, Aniruddha Kembhavi, Tanmay Gupta
**NeurIPS 2023**
Paper Project Code

6. **Improving Equivariance in State-of-the-Art Supervised Depth and Normal Predictors**
Yuanyi Zhong, **Anand Bhattad**, Yuxiong Wang, D.A. Forsyth
**ICCV 2023**
Paper Code

5. **Cut-and-Paste Object Insertion by Enabling Deep Image Prior for Reshading**
**Anand Bhattad**, D.A. Forsyth
**3DV 2022** (**Spotlight presentation**)
Paper Project Code

4. **DIVeR: Real-time and Accurate Neural Radiance Fields with Deterministic Integration for Volume Rendering**
   Liwen Wu, Jae Yong Lee, **Anand Bhattad**, Yuxiong Wang, D.A. Forsyth
   **CVPR 2022** (**Best Paper Finalist**; **Oral presentation**)
   Paper Project Code
3. **View Generalization for Single Image Textured 3D Models**
   **Anand Bhattad**, Aysegul Dundar, Guilin Liu, Andrew Tao, Bryan Catanzaro
   **CVPR 2021**
   Paper Project
2. **Unrestricted Adversarial Perturbations via Semantic Manipulation**
   **Anand Bhattad***, Min-Jin Chong*, Kaizhao Liang, Bo Li, D.A. Forsyth
   **ICLR 2020**
   Paper Code
1. **Improved Style Transfer with Calibrated Metrics**
   Mao-Chuang Yeh*, Shuai Tang*, **Anand Bhattad**, Chuhang Zou, D.A. Forsyth
   **WACV 2020**
   Paper

## Refereed Workshop Papers

5. **Intrinsic LoRA: A Generalist Approach for Discovering Knowledge in Generative Models**
   Xiaodan Du, Nicholas Kolkin, Greg Shakhnarovich, **Anand Bhattad**
   **CVPR-W 2024** (**Oral presentation**)
   workshops: Generative Models for CV, AI for 3D Gen, SyntaGen, SynthData4CV, Dataset Distillation
   Paper Project Code
4. **UrbanIR: Large-Scale Urban Scene Inverse Rendering from a Single Video**
   Zhi-Hao Lin, Bohan Liu, Yi-Ting. Chen, D.A. Forsyth, Jia-Bin Huang, **Anand Bhattad**, Shenlong. Wang
   **CVPR-W 2024**
   workshop: SynthData4CV
   Paper Project Code
3. **MIMIC: Masked Image Modeling with Image Correspondences**
   Kalyani Marathe, Mahtab Bigverdi, Nishat Khan, Tuhin Kundu, Patrick Howe, Sharan Ranjit S, **Anand Bhattad**, Aniruddha Kembhavi, Linda G. Shapiro, Ranjay Krishna
   **CVPR-W** 2024
   workshop: 3D with Multi-View Supervision (Archival long paper)
   Paper Code
2. **Big but Imperceptible Adversarial Perturbations via Semantic Manipulation**
   **Anand Bhattad***, Min-Jin Chong*, Kaizhao Liang, Bo Li, D.A. Forsyth
   **CVPR-W 2019** (**Oral presentation**)
   workshop: Adversarial Machine Learning in Real-World Computer Vision Systems
   Paper Code
1. **Detecting Anomalous Faces with "No Peeking" Autoencoders**
   **Anand Bhattad**, Jason Rock, D.A. Forsyth
   **CVPR-W 2018** (**Oral presentation**)
   workshop: Vision with Biased and Scarce Data
   Paper

## Technical Reports

4. **Generative Models: What Do They Know? Do They Know Things? Let's Find Out!**
   Xiaodan Du, Nicholas Kolkin, Greg Shakhnarovich, **Anand Bhattad**
   Technical Report, arXiv 2024
   Paper Project Code

3. **Blocks2World: Controlling Realistic Scenes with Editable Primitives**
   Vaibhav Vavilala, Seemandhar Jain, Rahul Vasanth, **Anand Bhattad**, D.A. Forsyth
   Technical Report, arXiv 2023
   Paper

2. **Make It So: Steering StyleGAN for Any Image Inversion and Editing**
   **Anand Bhattad**, Viraj Shah, Derek Hoiem, D.A. Forsyth
   Technical Report, arXiv 2023
   Paper Project

1. **SIRfyN: Single Image Relighting from your Neighbors**
   D.A. Forsyth, **Anand Bhattad**, Pranav Asthana, Yuanyi Zhong, Yuxiong Wang
   Technical Report, arXiv 2021
   Paper

## Patents

1. **Intrinsic-ControlNet: Zero-shot Object Compositing from Image Intrinsics**
   Zitian Zhang, Frédéric Fortier-Chouinard, Mathieu Garon, **Anand Bhattad**, Jean-François Lalonde
   Pending

## Advising

### Postdoctoral Researchers

- **Jamila Taaki** (JHU, primary advisor Laurent Pueyo) 2026 – now

### Doctoral Students

- **Deming (Remus) Li** (JHU, co-advised with Rama Chellappa) 2026 – now
- **Jiahan Zhang** (JHU) 2026 – now
- **Trong-Tung Nguyen** (JHU) 2026 – now

### Master's Students

- **Chaohao Yang** (JHU) 2026 – now
- **Joanna Cheng** (JHU, co-advised with Homanga Bharadhwaj) 2026 – now
- **Jiafeng Gu** (JHU) 2026 – now
- **Rayhaneh Behravesh** (JHU) 2026 – now
- **Yixin (Tracy) Zhu** (UChicago); Project on Improving Projective Geometry 2024 – 2025
- **Feiran Wang** (UIUC); Project on NeRF + Latent Illumination 2023
- **Kexuan (Klaus) Zou** (UIUC | Now: Software Engineer at NVIDIA); Project on 2D Meshes 2019 – 2020

### Undergraduate Students

- **Jin Hong (Jayden) Moon** (JHU) 2026 – now
- **Jiuzhou (Anthony) Cheng** (JHU) 2026 – now
- **Anudeep Chimata** (BITS Goa, India); Project on Representation Learning & Generative Models 2025 – now
- **Joshua Ahn** (UChicago); Project on Wavelets in Neural Radiance Fields 2024

- **Zhiyan (Alex) Wang** (UChicago); Project on Improving Projective Geometry 2024 – 2025
- **Kuan-Sheng Chen** (UIUC); Project on Improving Generative Models with Intrinsic Images 2023
- **Brian Chen** (UIUC | REU at CMU's RI | Next: Hive AI); Project on Lighting Correction 2020 – 2022
- **Anchu Zhu** (UIUC | Next: MS in CS at USC); Project on Anomaly Detection 2018 – 2019

## Student Collaborators

*Students whom I actively co-mentor or collaborate with (home institution in parentheses).*

### Doctoral-Level

- **Feng Wang** (JHU); **Papers in NeurIPS 2026** & under review 2026 – now
- **Namitha Guruprasad** (JHU) 2026 – now
- **Timing Yang** (JHU); **Papers in NeurIPS 2026** 2026 – now
- **Vivekjyoti Banerjee** (JHU); Paper under Review 2026 – now
- **Zhaoliang Zhang** (JHU) 2026 – now
- **Ozgur Kara** (UIUC) 2026 – now
- **David Serrano-Lozano** (UAB, Spain); **Paper in NeurIPS 2026** 2025 – now
- **Prakhar Kaushik** (JHU); **Paper in CVPR 2026** 2025 – 2026
- **Soumava Paul** (JHU); **Paper in CVPR 2026** 2025 – now
- **Tianchong Jiang** (TTIC); **Paper in ICRA 2026** (**Best Paper Award**) 2025 – 2026
- **Vinayak Gupta** (University of Maryland); **Paper in CVPR 2026** 2025 – 2026
- **Jun Myeong Choi** (UNC Chapel Hill); **Paper in CVPR 2025** 2024 – 2025
- **William Gao** (UChicago); **Paper in NeurIPS 2024** 2024 – now
- **Xiaoyan Xing** (University of Amsterdam); **Papers in CVPR 2025 & ICML 2026** 2024 – now
- **Ayush Sarkar** (UIUC); **Paper in CVPR 2024** 2023 – 2024
- **Kalyani Marathe** (University of Washington); **Paper in CVPR-W 2024** 2023 – 2024
- **Matthew Wallingford** (University of Washington); **Paper in NeurIPS 2024** 2023 – 2024
- **Vaibhav Vavilala** (UIUC); **Papers in ICLR 2026, 3DV 2026 & NeurIPS 2026** 2023 – 2025
- **Xiang Fan** (University of Washington); **Paper in ECCV 2024** 2023 – 2024
- **Xiao Zhang** (UChicago); **Papers in NeurIPS 2024 & ICML 2026** 2023 – 2025
- **Xiaodan Du** (TTIC); **Paper in CVPR-W 2024** 2023 – 2024
- **Zhi-Hao Lin** (UIUC); **Paper in CVPR-W 2024 & Paper in 3DV 2025** 2023 – 2024
- **Zitian Zhang** (Université Laval); **Papers in WACV 2025 & 3DV 2026** 2023 – 2024

### Master's-Level

- **Naveen Chandra R** (IISc, India); **Paper in NeurIPS 2026** 2026 – now
- **Shivam Tripathi** (IISc, India); **Papers in CVPR 2026 & NeurIPS 2026** 2025 – now
- **Varun Varma Thozhiyoor** (IISc, India); **Papers in CVPR 2026 & NeurIPS 2026** 2025 – now
- **Frédéric Fortier-Chouinard** (Université Laval); **Papers in WACV 2025 & 3DV 2026** 2023 – 2025
- **Hanlin (Asher) Mai** (UIUC); **Paper in CVPR 2024** 2023 – 2025
- **James Soole** (UIUC | Now: Research Engineer at MATLAB); **Paper in CVPR 2024** 2023
- **Seemandhar Jain** (UIUC | Now: PhD at UCSD); **Papers in ICLR 2026, 3DV 2026 & NeurIPS 2024** 2023 – 2024
- **Pranav Asthana** (UIUC | Now: PhD at UMD); Project on Relighting from Neighbors 2021 – 2022

### Undergraduate-Level

- **Furkan Horoz** (METU, Turkey / UIUC) 2026 – now
- **Pradhaan S Bhat** (PES University / IISc, India); **Paper in NeurIPS 2026** 2026 – now
- **Amitabh Mahapatra** (UIUC | Now: PhD at UIUC); **Paper in CVPR 2024** 2023 – 2025

- **Oscar Michel** (AI2 | Now: PhD at NYU); **Paper in NeurIPS 2023** 2023
- **Liwen Wu** (UIUC | Now: PhD at UCSD); **Paper in CVPR 2022 (Best Paper Finalist)** 2022

### Doctoral Thesis Committees

- **William Gao**, "Neural Methods for 3D Mesh Editing" (UChicago) 2025
- **Xiao Zhang**, "Representation Learning from and for Generative Models" (UChicago) 2025
- **Xin Yuan**, "Interpretable Unsupervised Generative Learning via Factorized Architectures and Structured Bottlenecks" (UChicago) 2024

## Courses Designed

**Generative Vision: The Art and Science of Visual Synthesis**, Johns Hopkins University; *Course Instructor* Fall 2026

- Developed a new course on visual synthesis with generative models

**Learning-based Vision**, Johns Hopkins University; *Course Instructor* Spring 2026

- Developed a new course on modern approaches in computer vision
- Taught a class of 35 students from Johns Hopkins University

**Past Meets Present: A Tale of Two Visions**, TTIC; *Course Instructor* Spring 2024

- Developed a new course from scratch on Computer Vision, combining historical and modern approaches
- Taught a class of 20+ students from TTIC and UChicago

## Teaching Assistant

**Computer Science**, UIUC; *Graduate Teaching Assistant* 2016 – 2018

- Applied Machine Learning (CS498 AML), Fall 2018
- Data Structures (CS 225), Spring 2017
- Intro Computing: Engrg & Sci (CS 101), Spring 2016 & Fall 2016

## Invited Talks

**The Three P's of Modern Computer Vision: Pixels, Perception, and Physics**

- University of Maryland, Vision Seminar Oct 2026
- Indian Institute of Science, Bangalore, Kotak-IISC AI/ML Seminar Jan 2026
- Johns Hopkins University, Guest Lecture in Rama Chellappa's Machine Perception Course Nov 2025
- Johns Hopkins University, Physics of Learning Seminar, invited by Brice Ménard Oct 2025
- Johns Hopkins University, ECE Department Seminar Sep 2025

**Principia: What Video Models Know about High-School Physics**

- CVPR AC Workshop (Lightning Talk) Jun 2026

**Emergent Latent Intrinsic Representations for Scene Relighting**

- UChicago, Great Lakes Graphics Workshop Apr 2025

**What Generative Image Models Understand (and Don't) about the Physical World**

- Univ of Wisconsin at Madison Apr 2025
- Johns Hopkins University, CS/LCSR Seminar Mar 2025
- UC Irvine, CS Seminar Mar 2025
- Stony Brook University, CS Seminar Mar 2025
- UIUC: External Speaker Series Feb 2025
- ULaval: Vision Seminar Feb 2025
- UCSD: Pixel Cafe Seminar Jan 2025

**Are Generative Image Models Physically Grounded?**
- UPenn: GRASP Seminar — Dec 2024
- NYU: Guest Lecture in Saining Xie's Computer Vision Course — Dec 2024

**Generative Models Inside Out**
- Midwest Computer Vision Workshop, Indiana University — Sep 2024

**What Do Generative Image Models Know?**
- IIIT Hyderabad, India; Vision Seminar — Jan 2024
- Exactech, Inc.; Tech Talk — Oct 2023

**Exploring Knowledge in Generative Models**
- Stanford University in Jiajun Wu's group — Jun 2023
- University of Tubingen, Autonomous Vision Group — May 2023

**What Do Generative Models Know about Geometry and Illumination?**
- UC Berkeley: Vision Seminar — Apr 2023
- NVIDIA Research — Apr 2023
- MIT: Vision and Graphics Seminar — Apr 2023
- CMU: VASC Seminar — Mar 2023
- UW: RAIVN Vision Seminar — Mar 2023

**Towards a Productive and Fun PhD Experience**
- UIUC: Computer Vision Workshop — Apr 2023

**Learning about Light Without Labeled Data**
- UMD: Vision Seminar — Mar 2023
- UCSD: Pixel Cafe Seminar — Feb 2023
- TTIC: Research Talk — Feb 2023

**Drag-and-Drop Rendering: Towards In the Wild Image Editing**
- CMU: Misc-Read Vision Reading Group — Aug 2021

## Services & Leadership

### Reviewing
- **Area Chair**, ICLR 2027, CVPR 2027, NeurIPS 2026, CVPR 2026, ICLR 2026, 3DV 2026, NeurIPS 2025, ICCV 2025, CVPR 2025, WACV 2025 — 2024 – now
- **Reviewer**, CVPR, NeurIPS, ECCV, ICCV, ICLR, SIGGRAPH, SIGGRAPH Asia and PAMI — 2018 – now

### University Service
- **Faculty Hiring Committee**, Department of Computer Science, Johns Hopkins University — 2025 – 2026

### Community Building Workshops
- **Lead Organizer**, "Bitter Lessons" workshop at CVPR — Jun 2026
- **Lead Organizer**, "How to Stand Out in the Crowd?" workshop at CVPR — Jun 2025
- **Lead Organizer**, "CV 20/20: A Retrospective Vision" workshop at CVPR — Jun 2024
- **Lead Organizer**, "Scholars & Big Models: How Can Academics Adapt?" workshop at CVPR — Jun 2023

### Technical Workshops
- **Organizer**, Findings, a new track at CVPR — Jun 2026
- **Organizer**, "Humans of Generative Models (HuGs)" at CVPR — Jun 2026

- **Lead Organizer**, “UniLight Workshop” at ICCV Oct 2025
- **Co-Organizer**, “Ind3D: Enforcing geometric, physical, topological, and functional inductive bias in 3D generation” workshop at CVPR Jun 2025
- **Lead Organizer**, “Knowledge in Generative Models” workshop at ECCV Sep 2024
- **Co-lead Organizer**, “Multimodal Artificial Intelligence ” workshop at TTIC Aug 2024
- **Lead Organizer**, Computer Vision workshop at Allerton, UIUC Apr 2023
- **Committee**, “Adversarial ML in Real-World Computer Vision Systems", workshop at CVPR Jun 2019
- **Committee**, “Security and Privacy in Machine Learning" workshop at ICML Jun 2019

## Seminars and Reading Groups

- **Organizer**, UIUC Vision Seminar, a new weekly seminars inviting speakers outside UIUC 2021 – 2023
- **Organizer**, Vision Lunch (computer vision reading group) 2021 – 2023
- **Organizer**, UIUC Summer Vision Coffee 2018

**Co-Founder**, SchoolEngg 2013 – 2015

- Forum for high-school students; providing firsthand knowledge of all engineering disciplines
- Renamed as *PrepLift*, an education counselling start-up in India

**Co-Founder**, American Society of Civil Engineers (ASCE) NITK Chapter 2013 – 2015

- Co-Founded the *first* ASCE student chapter of India
- Served as treasurer for one year and as a senior advisor for the second year
- Coordinated with board members, faculty, and outside organizations to facilitate 10+ events